

BIG DATA, BIG BIAS?

Mapping User-Centric Methods in the Social Sciences

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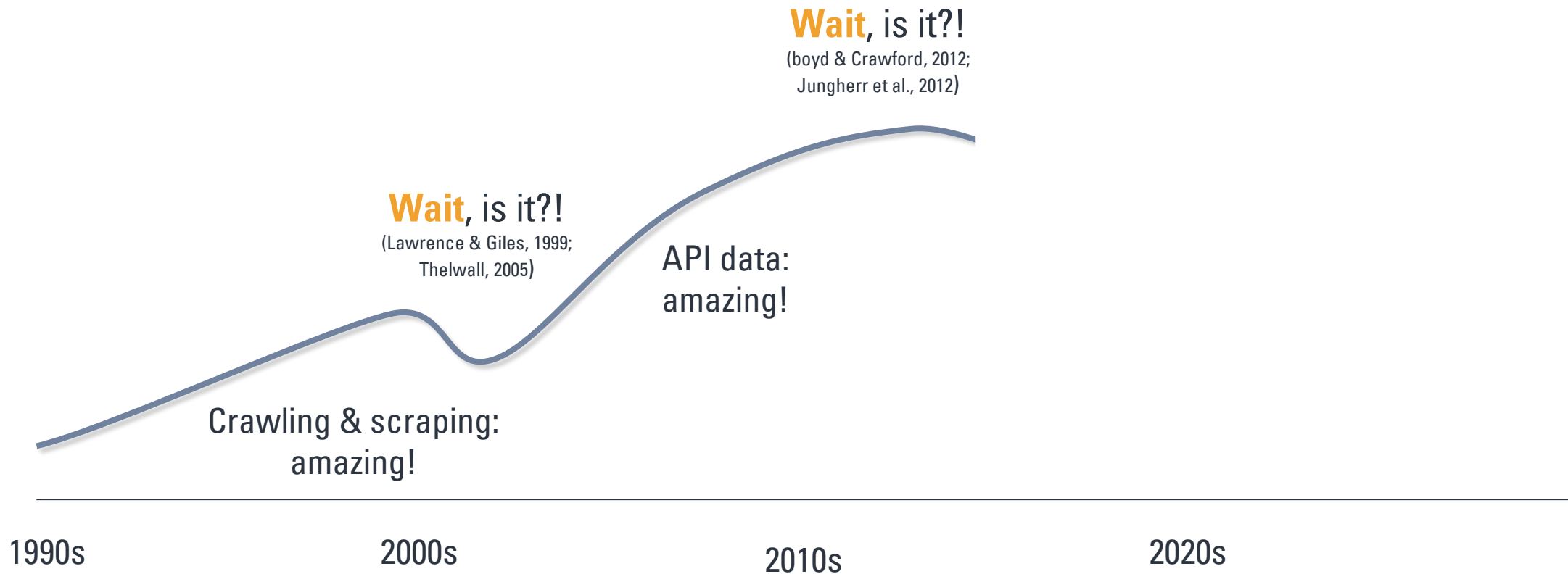
 [valeriehase](https://github.com/valeriehase)

 www.valerie-hase.com

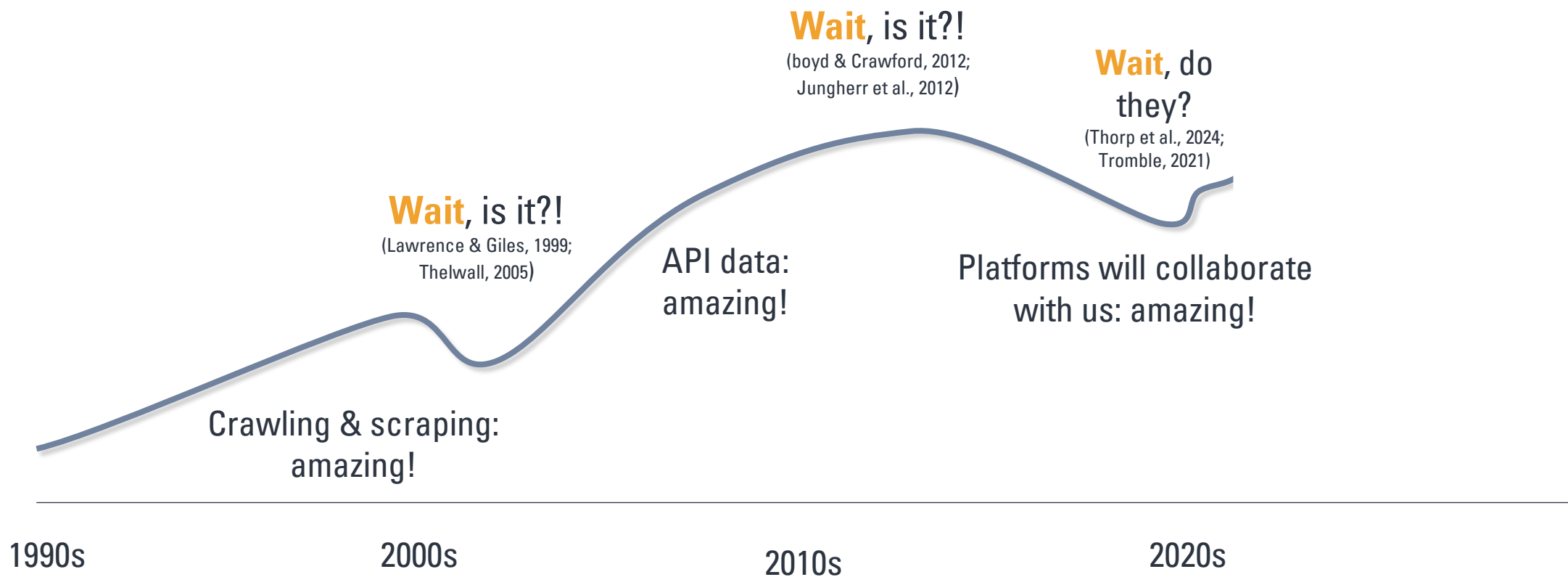
DIGITAL TRACE DATA: A TALE OVER TIME



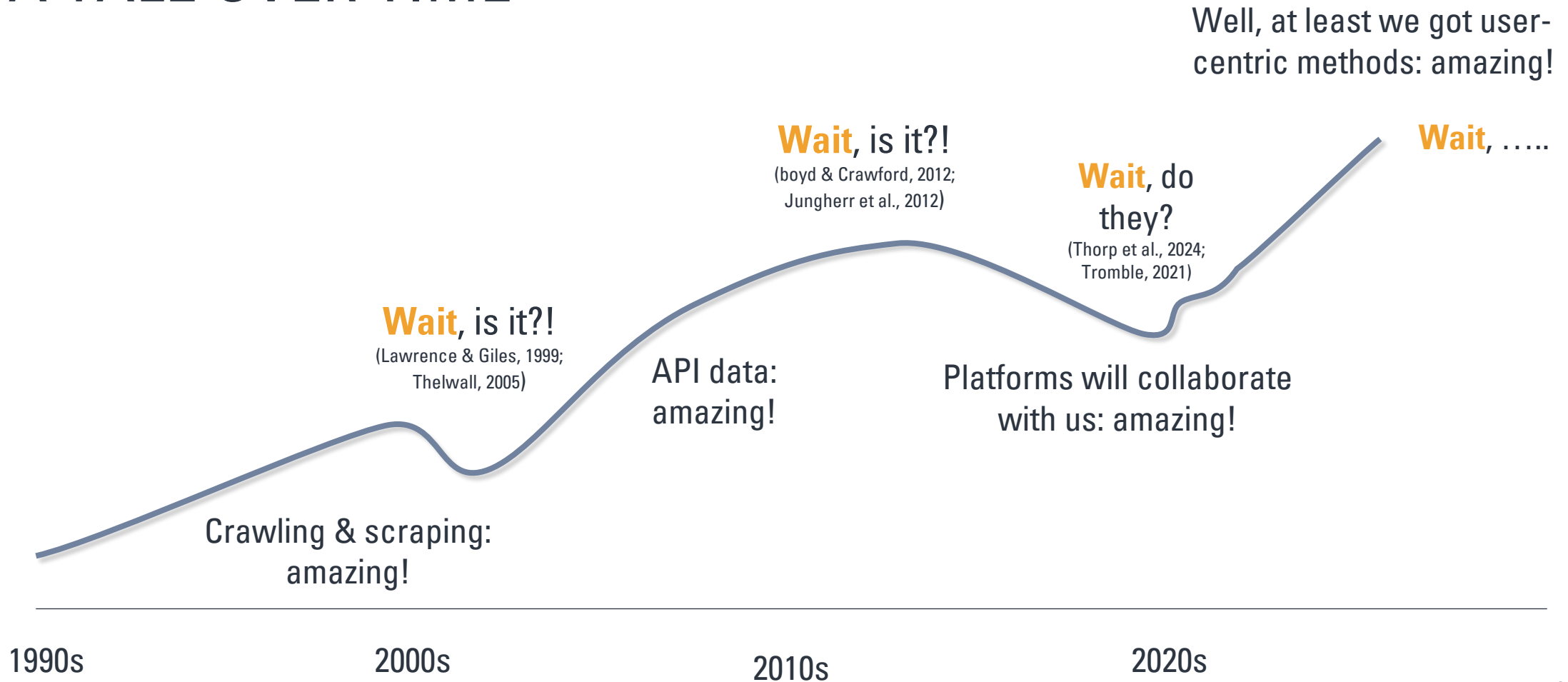
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KEY QUESTIONS



I. Define: What are user-centric methods?

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II. Map: How are they used in the social sciences?

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III. Improve: How do we move forward with them?

I. WHAT ARE USER-CENTRIC METHODS?

Digital trace data as “**records of activity** undertaken through an online information system” (Howison et al., 2011, p. 769).

Variety of user-centric methods for accessing trace data exist – including a **lack of conceptual agreement** on what these entail.



I. WHAT ARE USER-CENTRIC METHODS?

User-centric methods rely on participants' informed consent to collect their digital trace data



Passive Tracking

Participants are passively monitored
(e.g., via smartphone apps)



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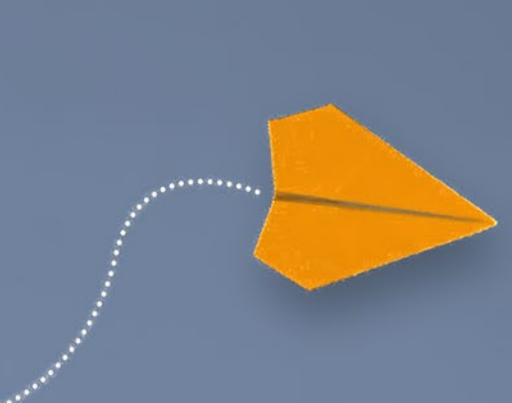
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Participants actively share data that others collected on them (e.g., via data download packages)



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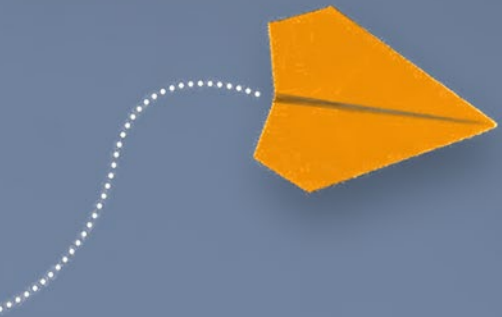
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Indiv. Identifiers

Participants share identifiers for databases with existing data (e.g., social media handles)



I. WHAT ARE USER-CENTRIC METHODS?

User-centric methods rely on participants' informed consent to collect their digital trace data

These methods (passive tracking, active sensors, data donation, and individual identifiers) differ in ...

- *which data they collect* (retro- vs. prospective, types of measures, researcher control)
- *when and how participants are involved* (steps, degree of burden)

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II. HOW ARE USER-CENTRIC METHODS USED IN THE SOCIAL SCIENCES?

While the use of trace data has increased since the mid-2010s (Lee & Peng, 2023; Shibuya, 2022), we know little about **how user-centric methods are used**.

- Reviews focus on specific methods and measures
(Shibuya et al., 2021; Parry et al., 2021)
- Studies collecting trace data may focus on the measurable rather than the relevant
(Wagner et al., 2021; Weiß et al., 2025)



II. HOW ARE USER-CENTRIC METHODS USED IN THE SOCIAL SCIENCES?

While the use of trace data has increased since the mid-2010s (Lee & Peng, 2023; Shibuya, 2022), we know little about **how user-centric methods are used**.

- **RQ1a:** *Which user-centric methods do researchers in the social sciences use to collect digital traces?*
- **RQ1b:** *For which measures do researchers in the social sciences empirically apply user-centric methods?*



II. HOW ARE USER-CENTRIC METHODS USED IN THE SOCIAL SCIENCES?

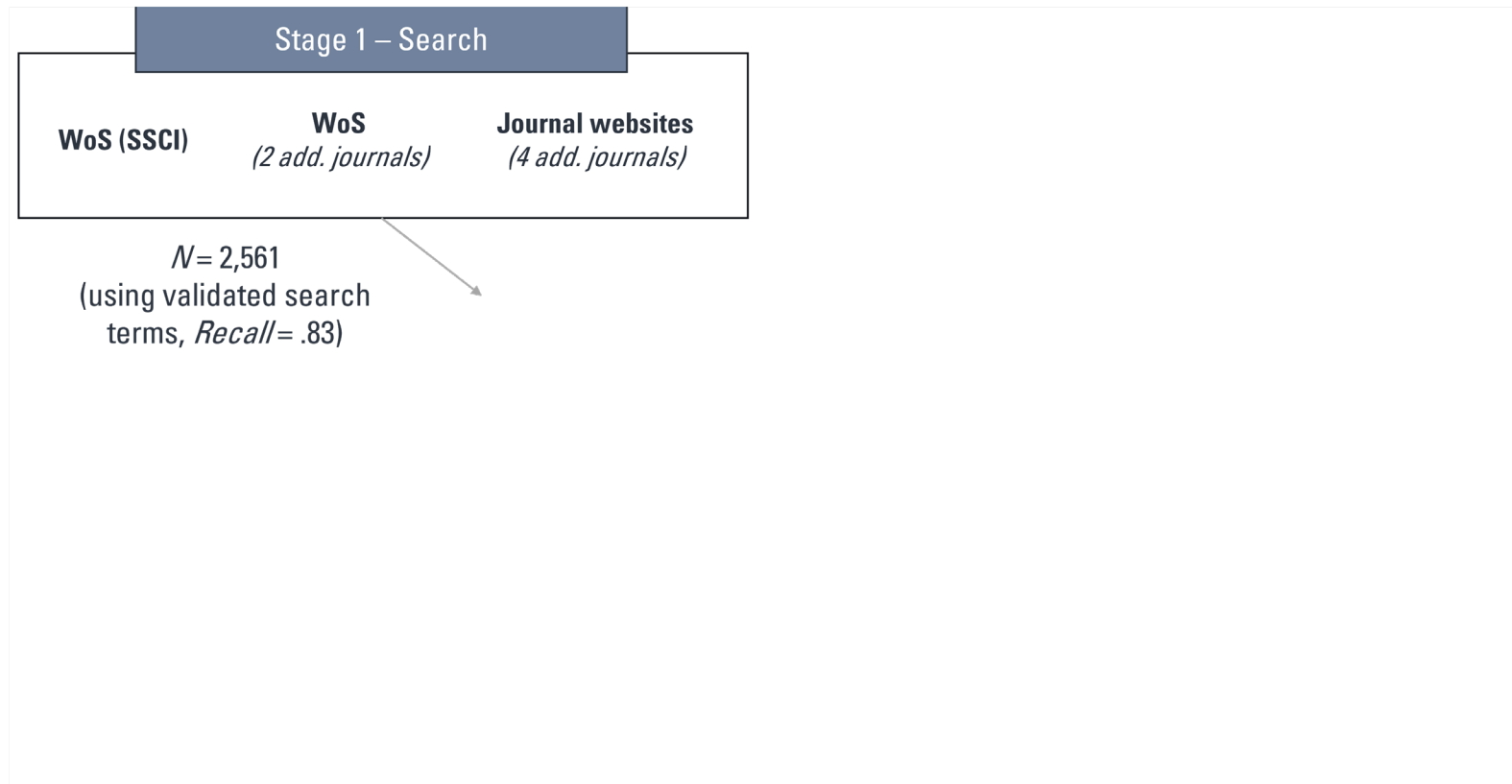
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- Preregistered, systematic literature review
- Focus on empirical or methodological work using digital traces in the social sciences (2015-2024)



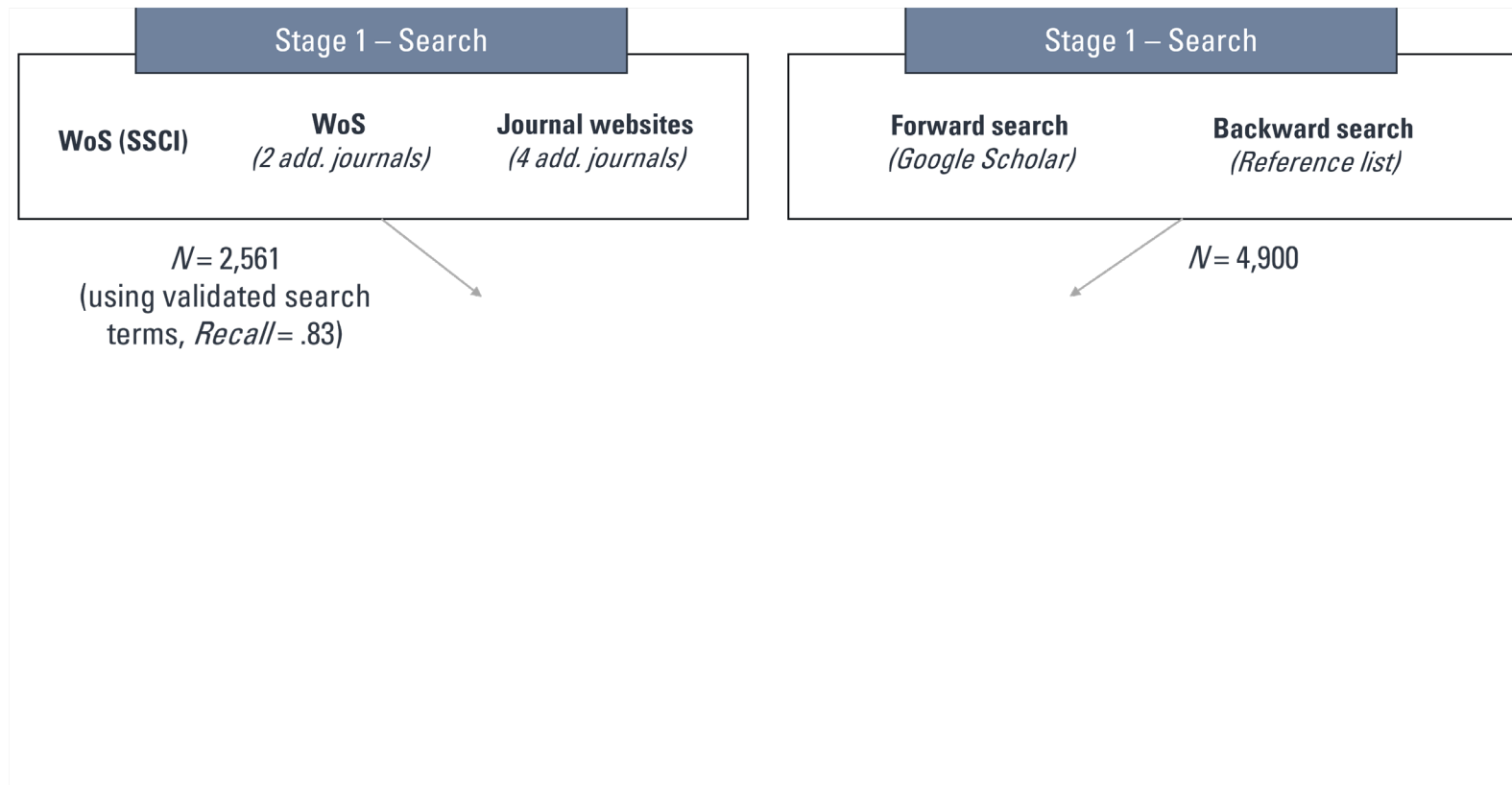
Preregistered, Systematic Literature Review

following PRISMA-2020 and STAMP guidelines (Page et al., 2019; Rogge et al., 2024)



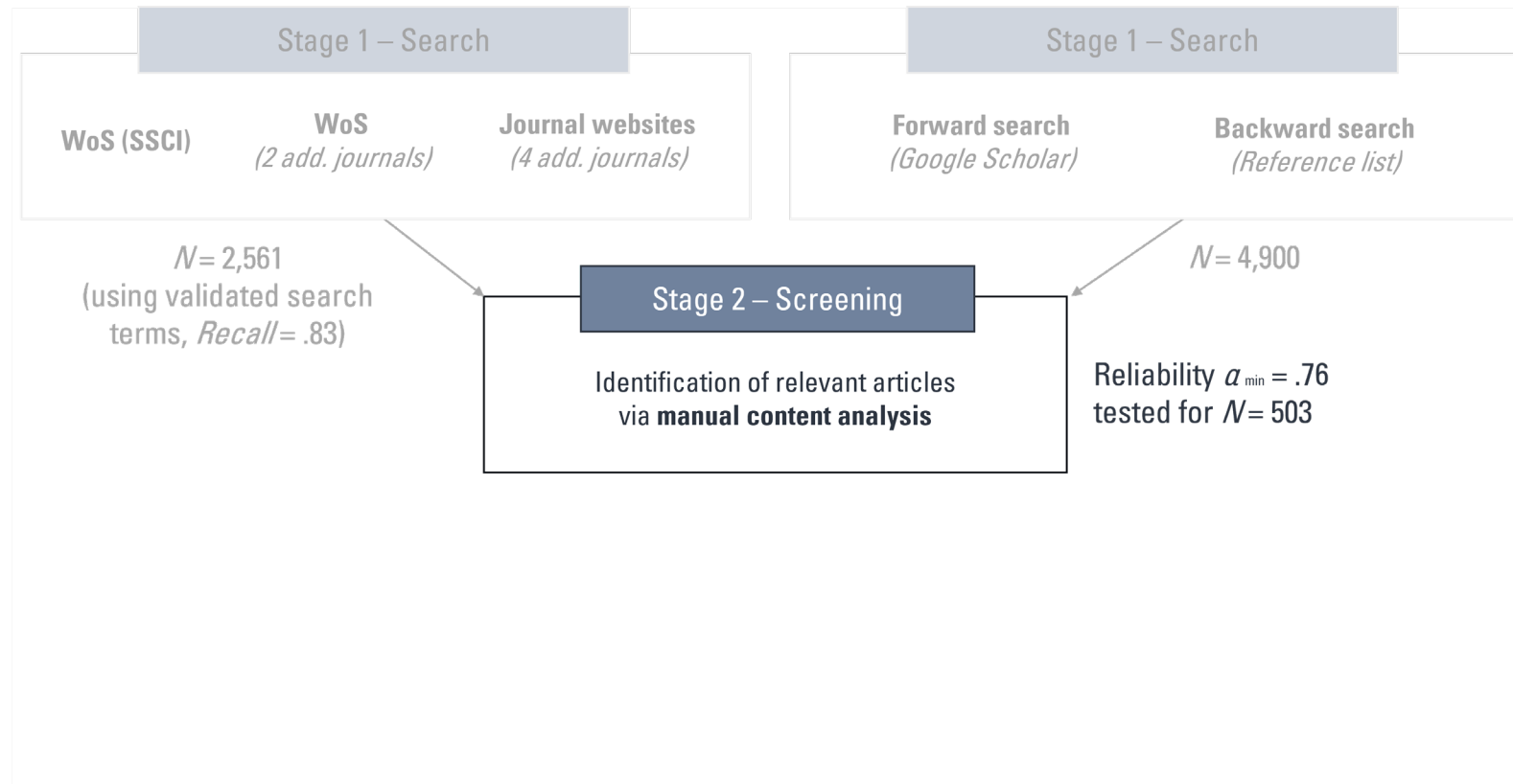
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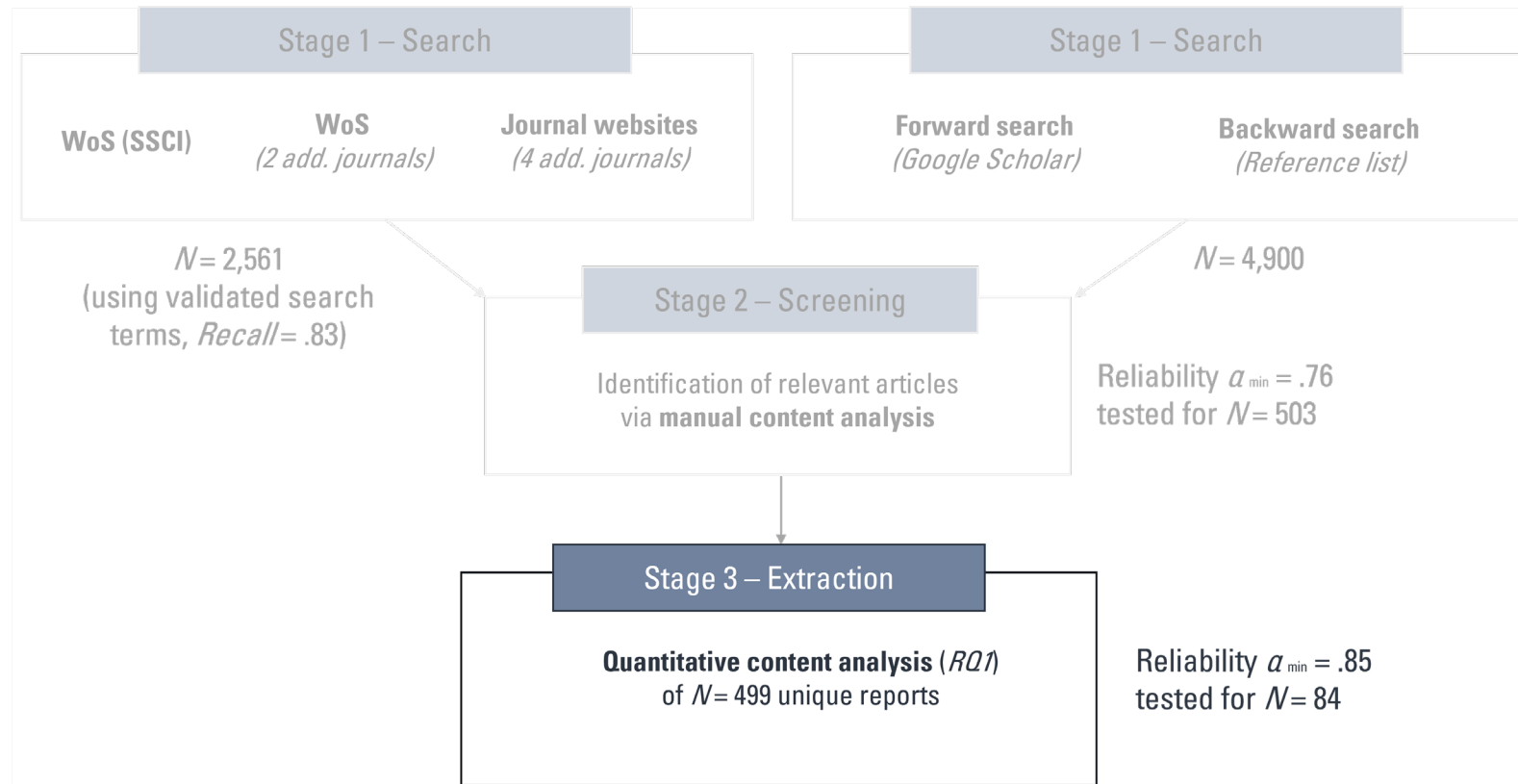
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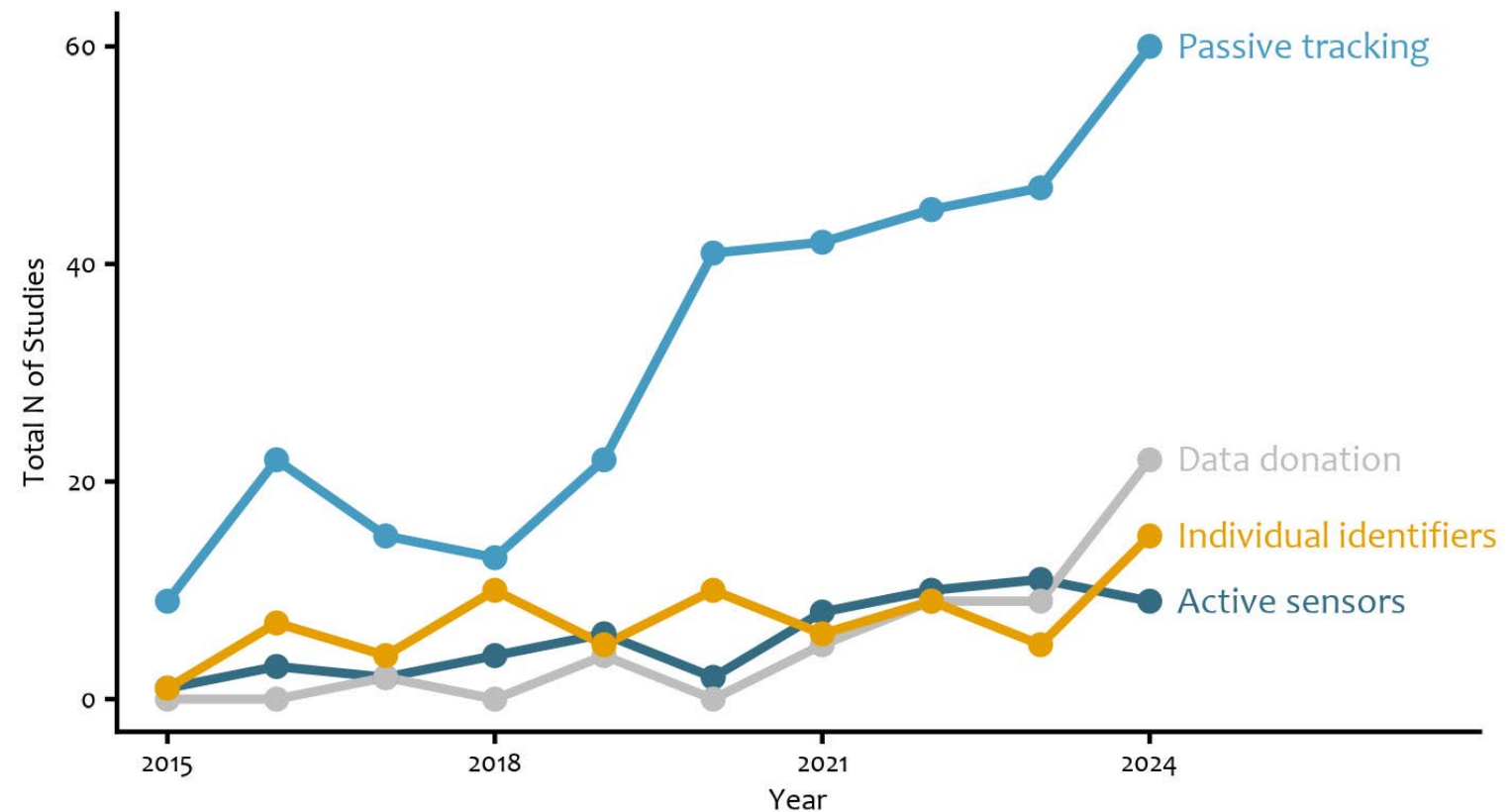




II. HOW ARE USER-CENTRIC METHODS USED IN THE SOCIAL SCIENCES?

Use of User-Centric Methods over Time

Number of studies per year and method

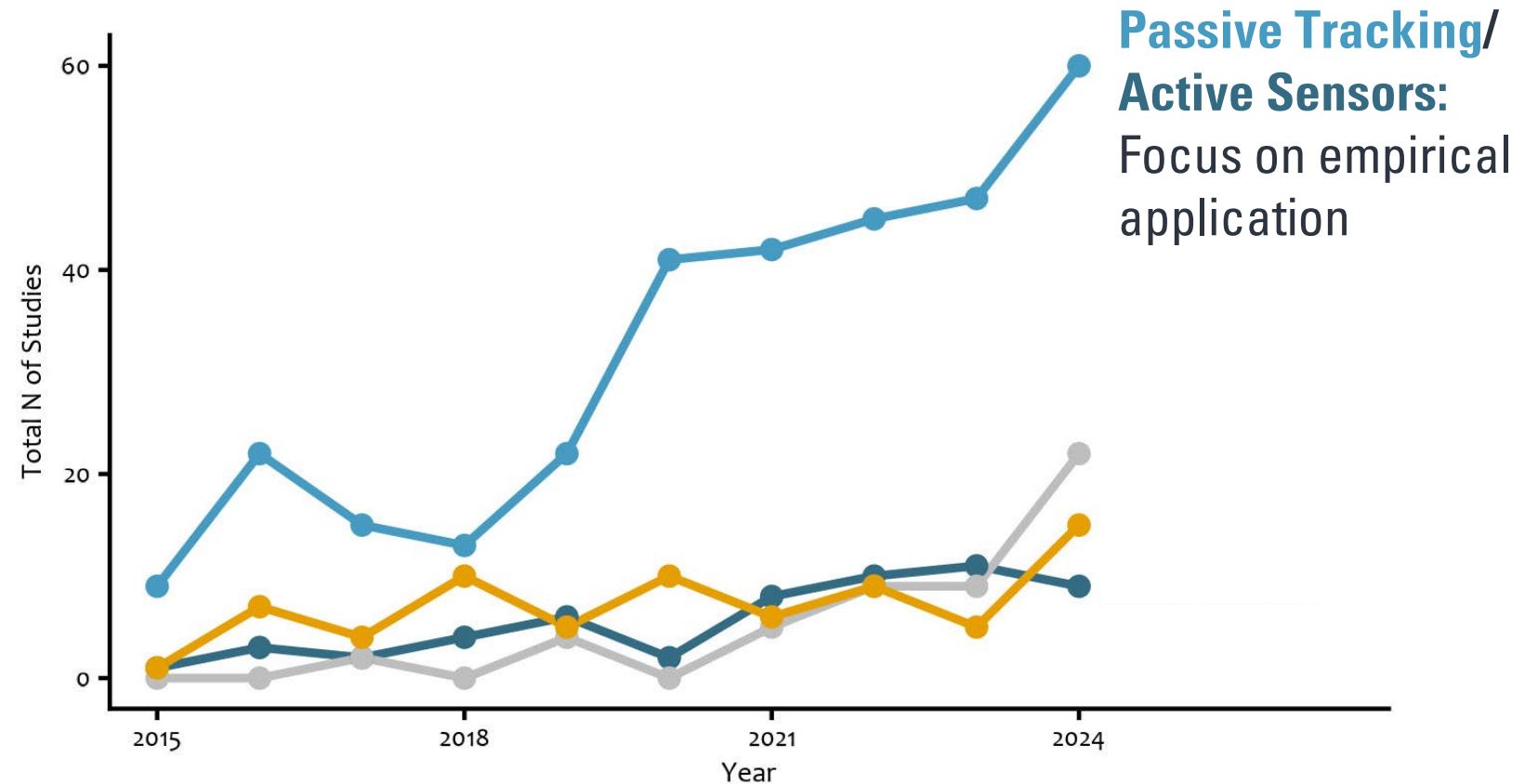




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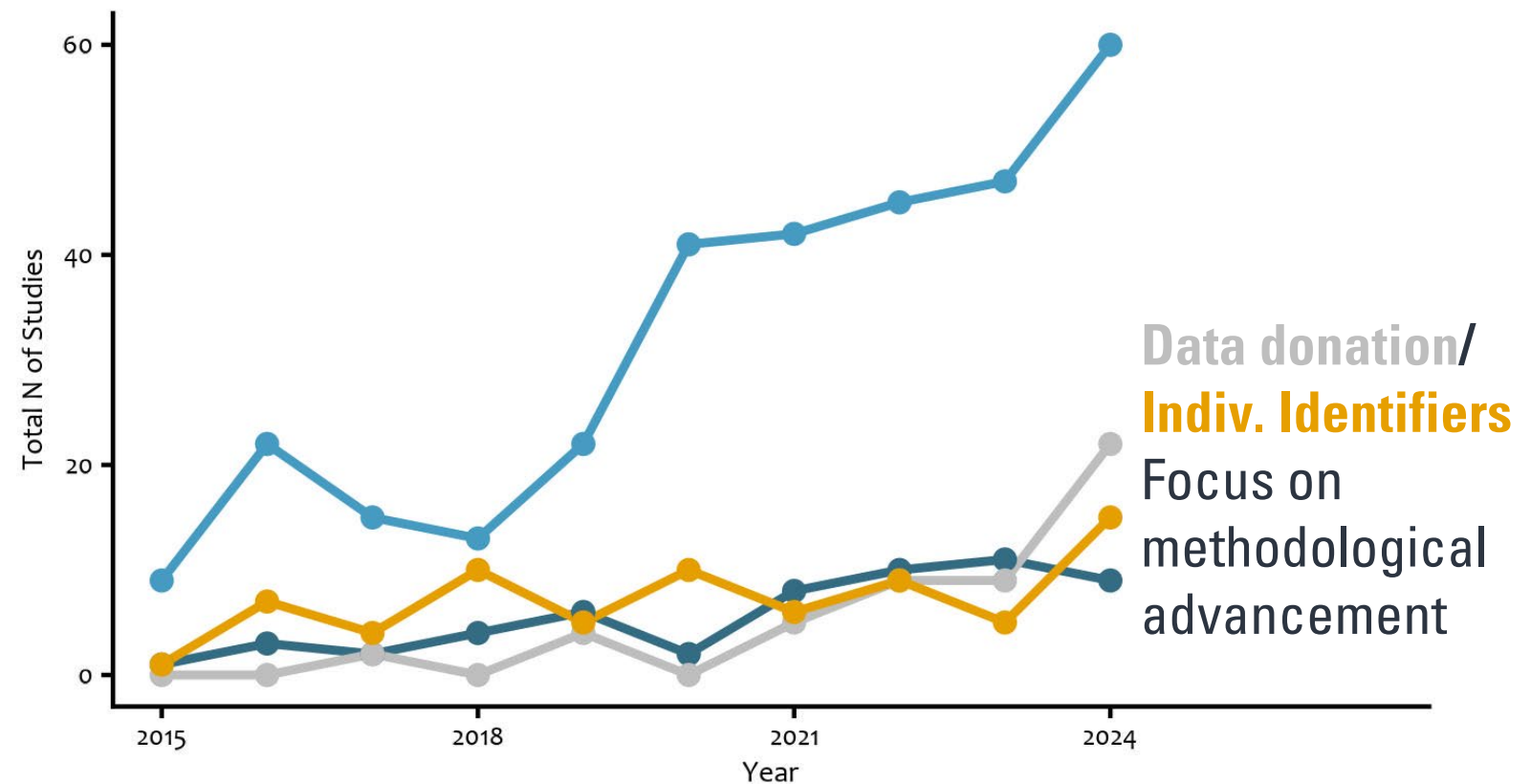




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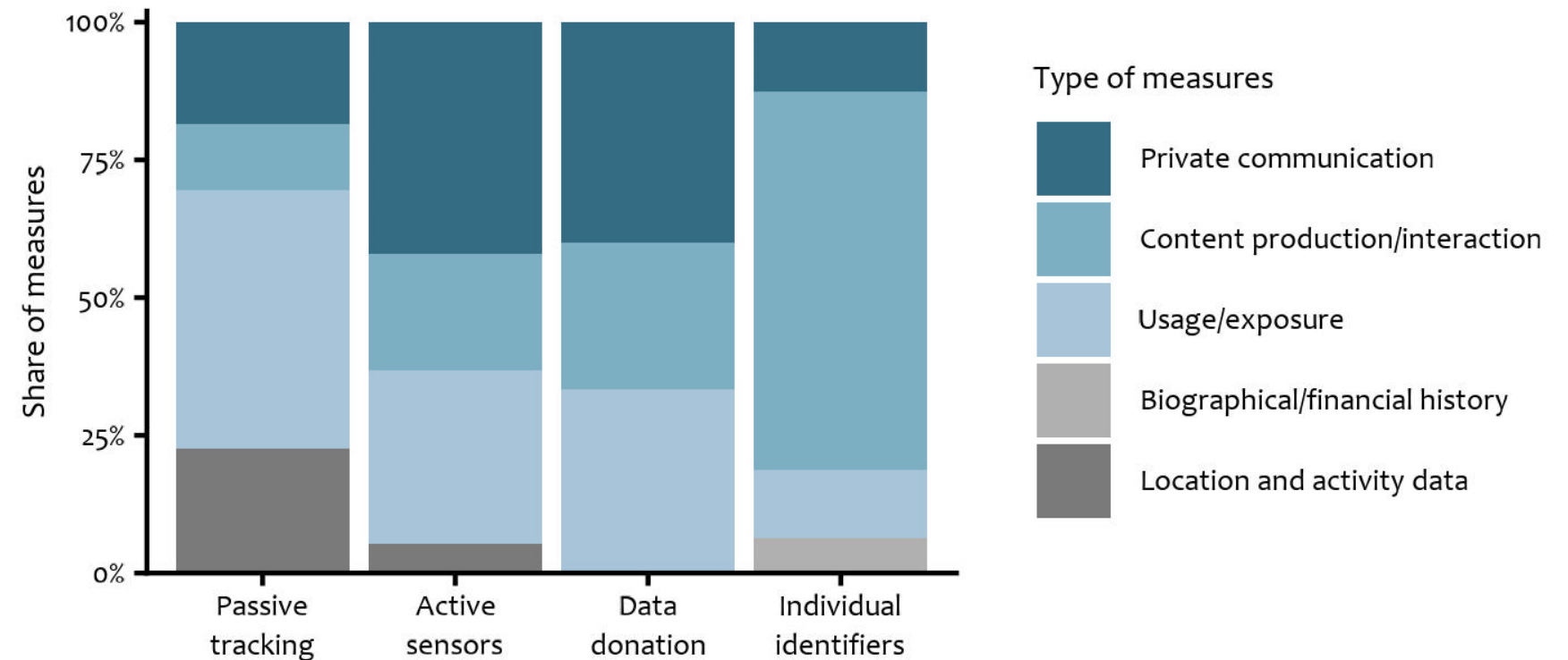
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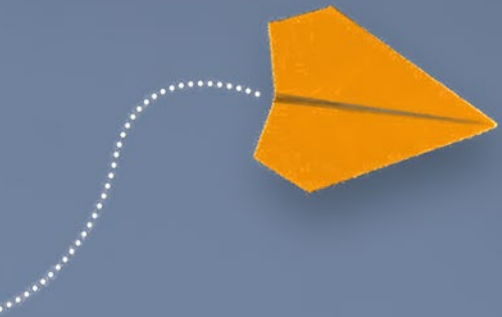


II. HOW ARE USER-CENTRIC METHODS USED IN THE SOCIAL SCIENCES?

Measures Collected across Methods

Share of measures per method (empirical studies)





II. HOW ARE USER-CENTRIC METHODS USED IN THE SOCIAL SCIENCES?

- Clear increase in use after API closures
- Passive tracking is (by far) the most popular method
- Communication researchers – often in interdisciplinary teams – shape method advancement (except for linkage via identifiers)
- Key methodological concerns remain (Wagner et al., 2021; Weiß et al., 2025):
 - Platform dependency in measurement construction
 - Limited data enrichment: “thin” over “thick” traces (Bornakker & Due, 2018)

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III. Improve: How do we move forward with them?

III. HOW CAN WE IMPROVE USER-CENTRIC METHODS?

By now, “the initial hype has levelled off”:

(Keusch & Kreuter, 2021, p. 112)

We need to... (Daikeler et al., 2024; Hase et al., 2025; Weiß et al., 2025)

- Conceptually **define bias** in digital trace data
- **Test** and **advance methods** for addressing bias



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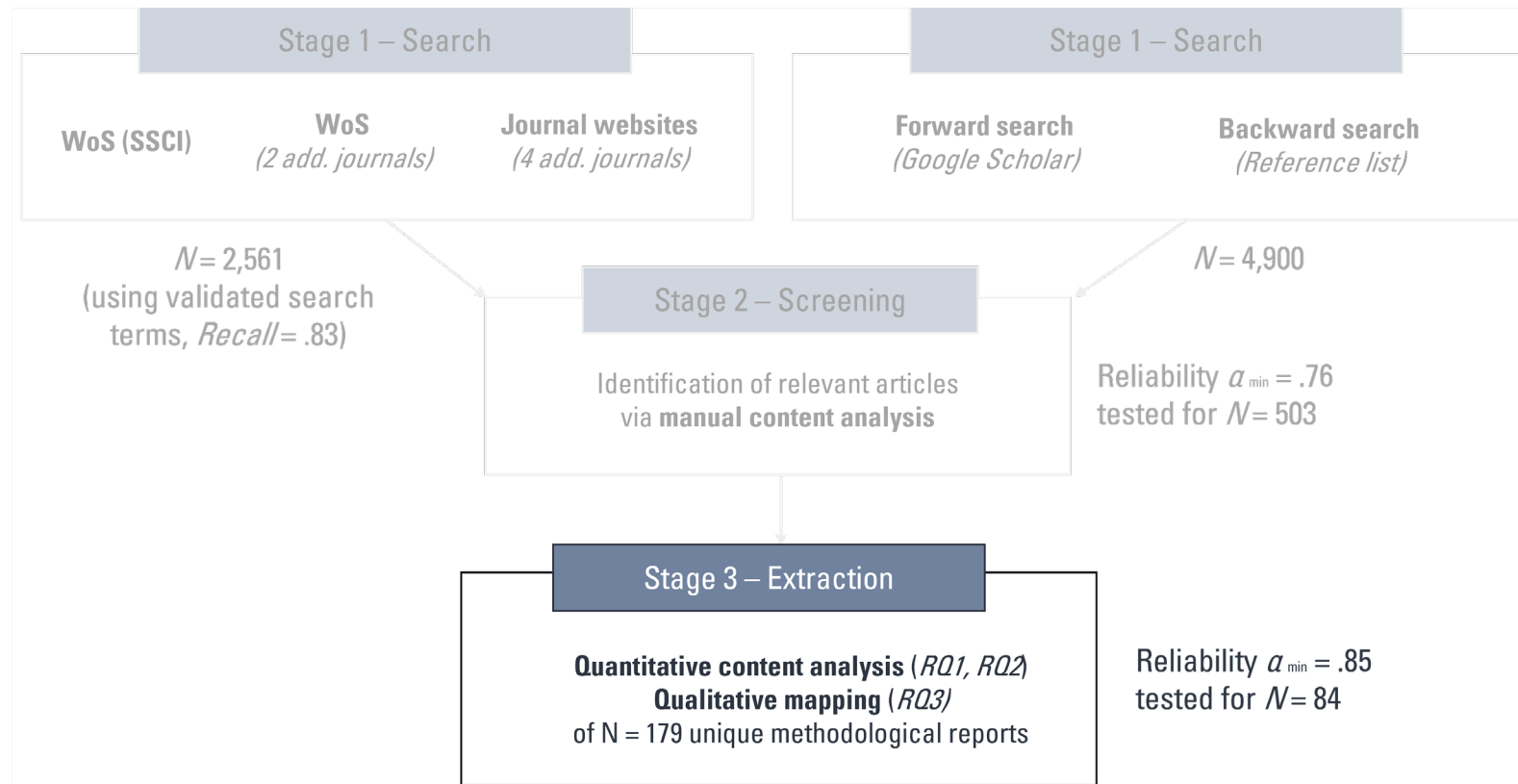
We need to... (Daikeler et al., 2024; Hase et al., 2025; Weiß et al., 2025)

- Conceptually **define bias** in digital trace data
- **Test** and **advance methods** for addressing bias
- **RQ2:** *How do researchers study bias in user-centric methods?*
- **RQ3:** *Which error correction methods exist?*

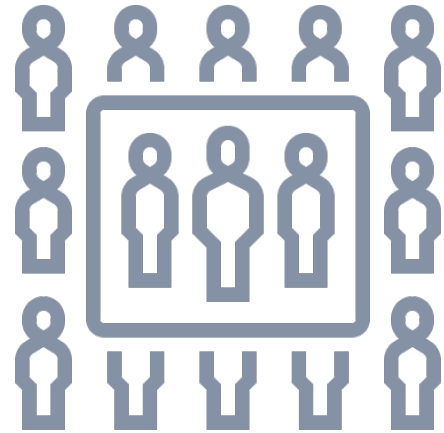


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III. HOW CAN WE IMPROVE USER-CENTRIC METHODS?



74 % of methodological studies:
Errors in representation



26 % of methodological studies: Measurement error



Focus on quantifying rather than
addressing error

III. HOW CAN WE IMPROVE USER-CENTRIC METHODS?



User-centric tool

Experience; Burden; Participant control

Survey design strategies

Interviewer; Incentives; Frequency, Placement or Framing of Invitation, Consent, Appeals, or Reminders

Collected Measures

Domain; Plattform

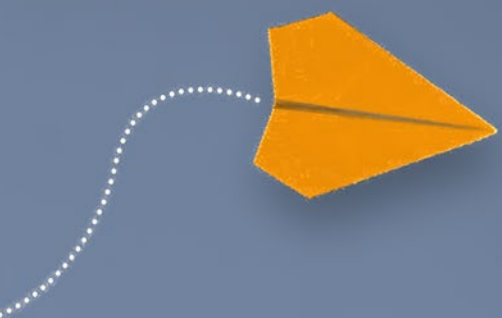
Statistical Error Correction

Weighting

III. HOW CAN WE IMPROVE USER-CENTRIC METHODS?



Comparison of survey and trace data
Correlation; Predictive Validity; MTMM
modelling



II. HOW ARE USER-CENTRIC METHODS USED IN THE SOCIAL SCIENCES?

- Stark increase in method development (tools, infrastructures), partly integrated in probability-based panels
- But:
 - We are more concerned about who gives us data than the quality of their data
 - Privacy-by-design (often) mean “thin” data by design: few approaches for data enrichment
 - Errors in representation: Traditional error correction methods (e.g., from survey research) don’t work
 - Measurement: We know little about when traces work “better”

CONCLUSION

- User-centric methods become more important
- Communication research at the forefront of developing these methods compared to, for example, text-as-data methods (Baden et al., 2023; Hase et al., 2022)

However,...

- We tend to (still) take trace data at face validity
 - Measures: Focus on the measurable rather than the relevant (Wagner et al., 2021), „thin“ data
 - Samples: „More“ instead of „better“ data
- For many methods we are, still, dependent on the good will of big tech companies



Integration rather than separation:
„Fit for purpose“ (Weiß et al., 2025)

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